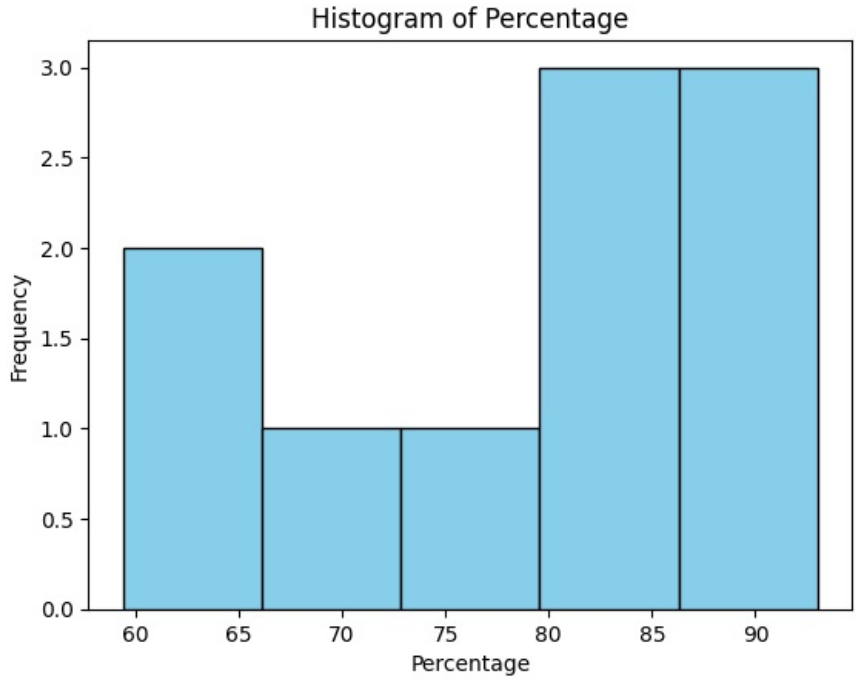
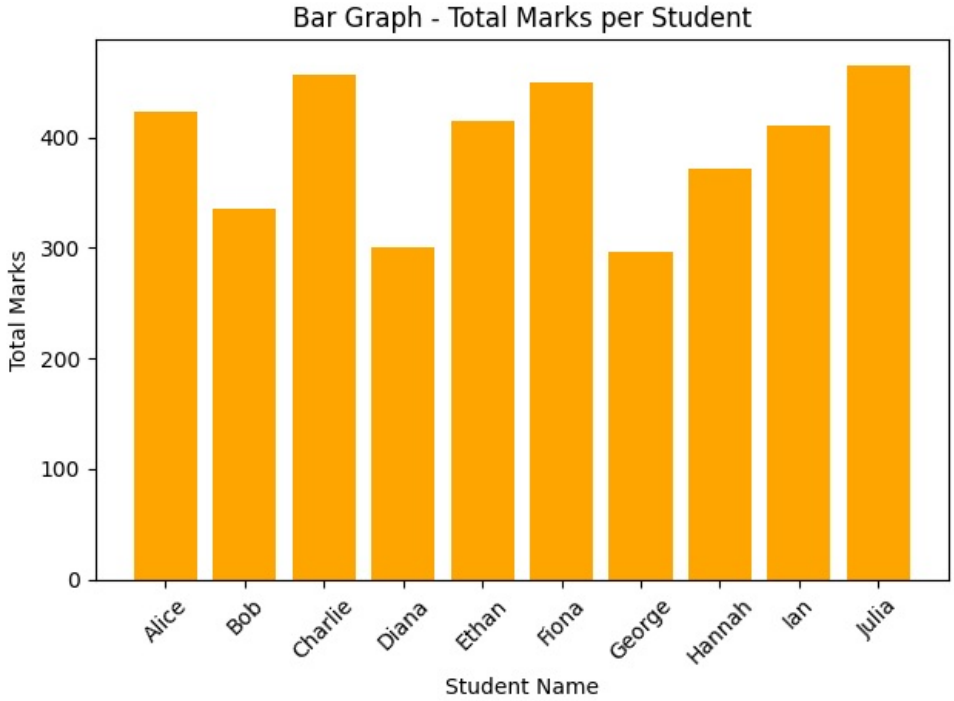
In [1]:
```python
import pandas as pd

# Sample data
data = {
    "Sr. No.": list(range(1, 11)),
    "Student Name": ["Alice", "Bob", "Charlie", "Diana", "Ethan", "Fiona", "George", "Hannah", "Ian", "Julia"],
    "Sub1": [78, 65, 95, 55, 80, 90, 60, 70, 85, 92],
    "Sub2": [85, 70, 92, 60, 78, 85, 58, 75, 80, 94],
    "Sub3": [90, 72, 88, 58, 85, 89, 55, 72, 79, 95],
    "Sub4": [82, 68, 91, 65, 88, 92, 63, 78, 82, 91],
    "Sub5": [88, 60, 90, 62, 84, 94, 61, 76, 84, 93],
}

df = pd.DataFrame(data)
df["Total"] = df[["Sub1", "Sub2", "Sub3", "Sub4", "Sub5"]].sum(axis=1)
df["Percentage"] = df["Total"] / 5
df["Grade"] = df["Percentage"].apply(lambda x: "A+" if x >= 90 else ("A" if x >= 80 else ("B+" if x >= 70 else

# Save to CSV
df.to_csv("students_data.csv", index=False)
```

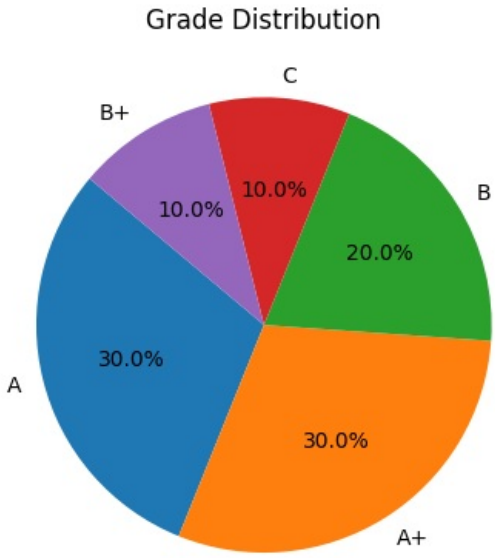
In [2]:
```python
import pandas as pd

df = pd.read_csv("students_data.csv")
print(df)
```

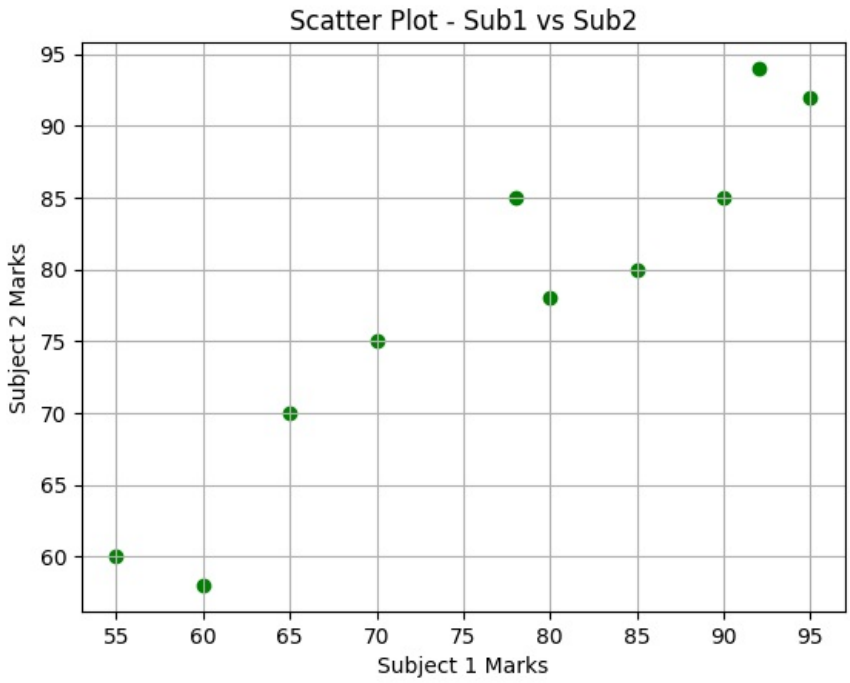
```
   Sr. No. Student Name  Sub1  Sub2  Sub3  Sub4  Sub5  Total  Percentage Grade
0        1        Alice    78    85    90    82    88    423        84.6     A
1        2          Bob    65    70    72    68    60    335        67.0     B
2        3      Charlie    95    92    88    91    90    456        91.2    A+
3        4        Diana    55    60    58    65    62    300        60.0     B
4        5        Ethan    80    78    85    88    84    415        83.0     A
5        6        Fiona    90    85    89    92    94    450        90.0    A+
6        7       George    60    58    55    63    61    297        59.4     C
7        8       Hannah    70    75    72    78    76    371        74.2    B+
8        9          Ian    85    80    79    82    84    410        82.0     A
9       10        Julia    92    94    95    91    93    465        93.0    A+
```

In [3]:
```python
import matplotlib.pyplot as plt

# Histogram of percentages
plt.hist(df["Percentage"], bins=5, color='skyblue', edgecolor='black')
plt.title("Histogram of Percentage")
plt.xlabel("Percentage")
plt.ylabel("Frequency")
plt.show()

# Bar Graph of total marks
plt.bar(df["Student Name"], df["Total"], color='orange')
plt.title("Bar Graph - Total Marks per Student")
plt.xlabel("Student Name")
plt.ylabel("Total Marks")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

# Pie chart of grades
grade_counts = df["Grade"].value_counts()
plt.pie(grade_counts, labels=grade_counts.index, autopct='%1.1f%%', startangle=140)
plt.title("Grade Distribution")
plt.show()

# Scatter plot - Sub1 vs Sub2
plt.scatter(df["Sub1"], df["Sub2"], color='green')
plt.title("Scatter Plot - Sub1 vs Sub2")
plt.xlabel("Subject 1 Marks")
plt.ylabel("Subject 2 Marks")
plt.grid(True)
plt.show()
```


Histogram of Percentage
Frequency
Percentage
0.0
0.5
1.0
1.5
2.0
2.5
3.0
60
65
70
75
80
85
90

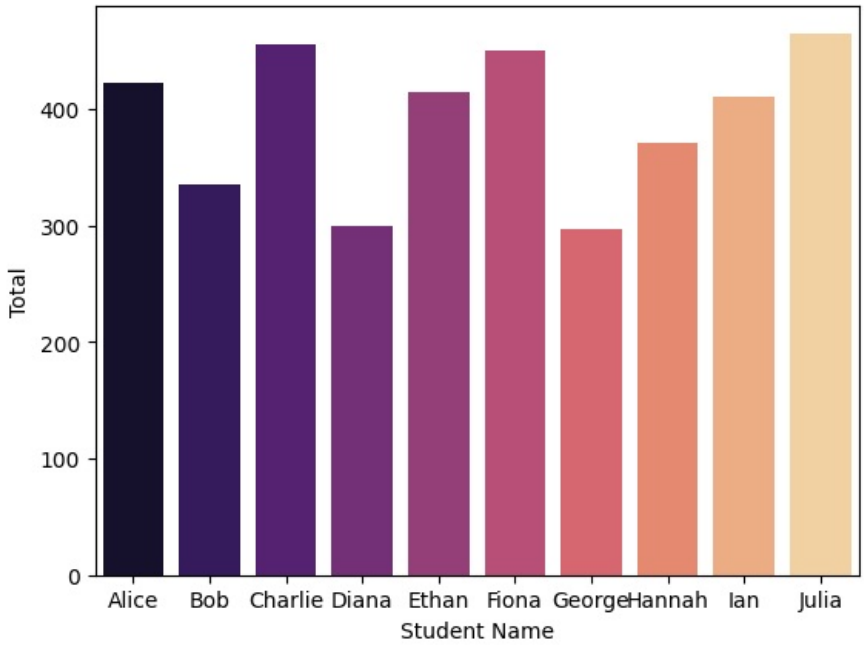

Bar Graph - Total Marks per Student
Total Marks
Student Name
0
100
200
300
400
Alice
Bob
Charlie
Diana
Ethan
Fiona
George
Hannah
Ian
Julia


Grade Distribution
C
B+
B
A
A+
10.0%
10.0%
20.0%
30.0%
30.0%



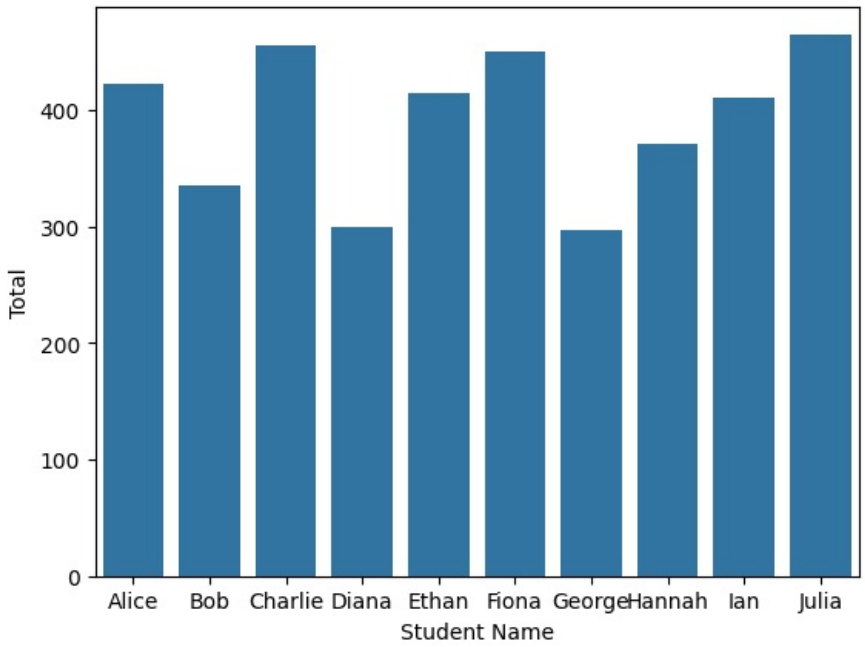
In [12]:
```
sns.barplot(x="Student Name", y="Total", data=df, hue="Student Name", palette="magma", legend=False)
```

Out[12]: <Axes: xlabel='Student Name', ylabel='Total'>



In [13]:
```
sns.barplot(x="Student Name", y="Total", data=df)
```

Out[13]: <Axes: xlabel='Student Name', ylabel='Total'>


Total
0
100
200
300
400
Alice
Bob
Charlie
Diana
Ethan
Fiona
George
Hannah
Ian
Julia
Student Name

In [ ]: